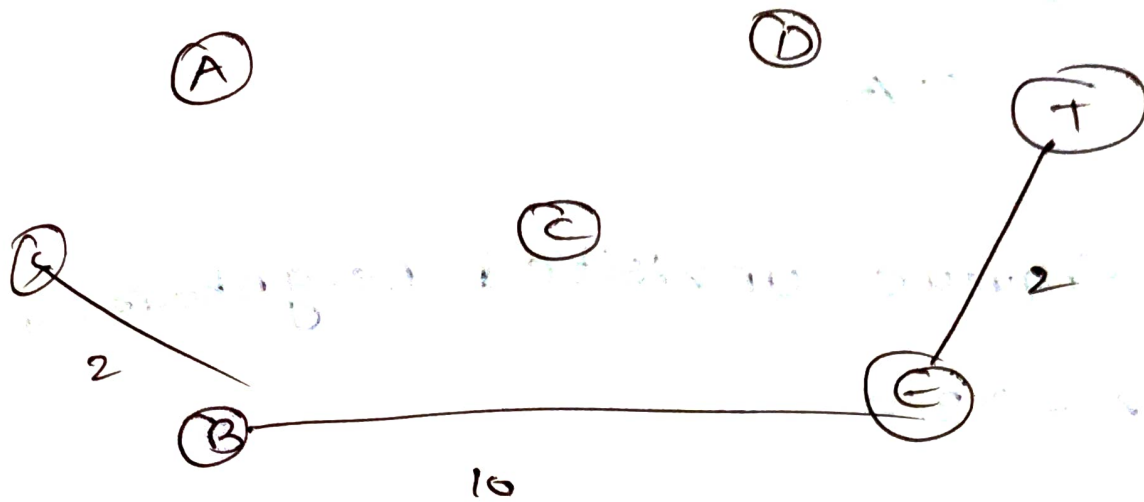
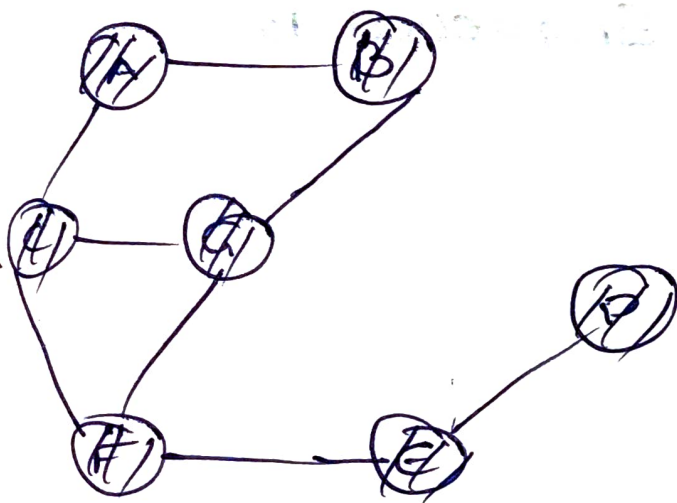


14 + 2 = 16



BFS



Visited

A	B	C	G	F	E	D
---	---	---	---	---	---	---

Adjacency List

A: []

B: [A, G]

C: [A, F, G]

D: [E]

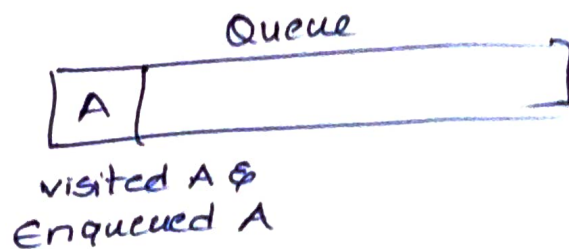
E: [D, F]

F: [C, E, G]

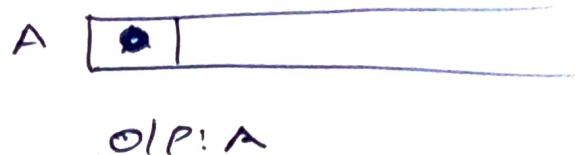
G: [B, C, F]

F: [C, E, G]

Step 1:



Step 2: Dequeue front



Step 3: Enqueue unvisited neighbours of

$A \rightarrow B, C$



Step 4: Visited B & ^{de}Enqueued B



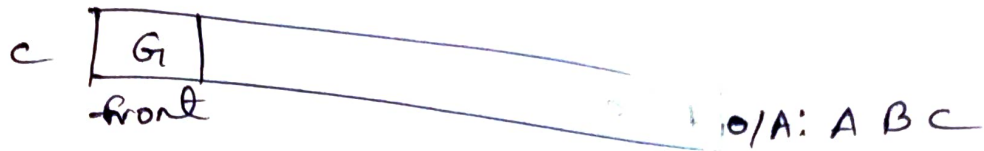
O/P: A B

Step 5: Enqueue unvisited neighbours of

$B \rightarrow G$

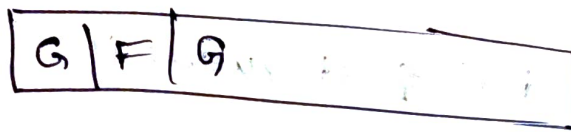


step 6: visited **c** & **c** is dequeued



step 7: Enqueue unvisited neighbours of

c $\rightarrow$ F, G



step 8: visited **G** & dequeued **G**

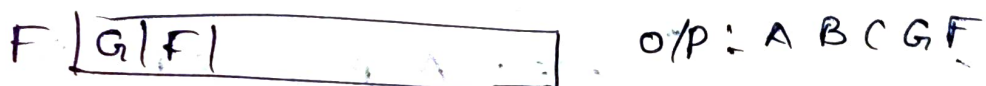


step 9: Enqueue unvisited neighbours of

G $\rightarrow$ F

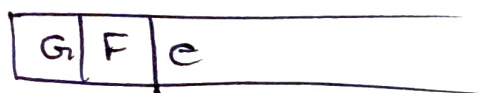


step 10: visited **F** & dequeued **F**



step 11: Enqueue unvisited neighbours of

F $\rightarrow$ e



step 12: since G is visited dequeue it

G | F | E

step 13: since F is visited dequeue it

F | E |

step 14: visited E & dequeue it

E |

O/P: A B C G F E

step 15: Enqueue unvisited neighbours of

E $\rightarrow$ D

D |

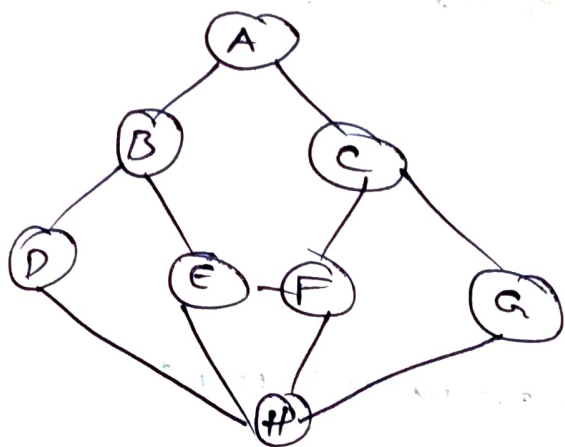
step 16: Visited D & dequeue ~~it~~ D

D |

O/P: A B C G F E D

step 17: enqueue unvisited / neighbours of
D $\rightarrow$

BFS



Adjacency List

A: [B, C]

B: [D, E, A]

C: [A, F, G]

D: [B, H]

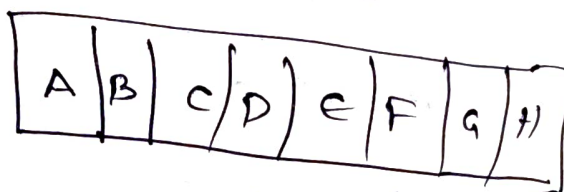
E: [B, F, H]

F: [C, E, H]

G: [C, H]

H: [E, F, D, G]

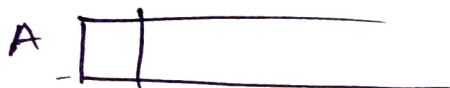
Visited



Step 1: Enqueue A



Step 2: Visited A & Dequeue front



O/p: A

step 3: Enqueue unvisited neighbours of

$A \rightarrow B, C$



step 4: visited B & Dequeue front



O/P: A B

step 5: Enqueue Unvisited neighbours of

$B \rightarrow D, E$



step 6: visited C Dequeue front



O/P: A B C

step 7: Enqueue unvisited neighbours

of $C \rightarrow F, G$



step 8: visited D & Dequeue front

D | E | F | G |

O/P: A B C D

step 9: Enqueue unvisited neighbours of D $\rightarrow$ E

E | F | G | H

step 10: visited E & dequeue front

E | F | G | H

O/P: A B C D E

step 11: Enqueue unvisited neighbours of E $\rightarrow$ F, H

E $\rightarrow$ F, H

F | G | H | F | H

step 12: Visited F & Dequeue front

F | G | H | F | H

O/P: A B C D E F

step 13 : Enqueue unvisited neighbours of F → H

G	H	F	H	H
---	---	---	---	---

step 14 : visited E dequeue front

G	H	F	H	H
---	---	---	---	---

O/P: A B C D E F G

step 15 : Enqueue unvisited neighbours of G → H

H	F	H	H	H
---	---	---	---	---

step 16 : ~~Dequeue~~ visited H & dequeue front

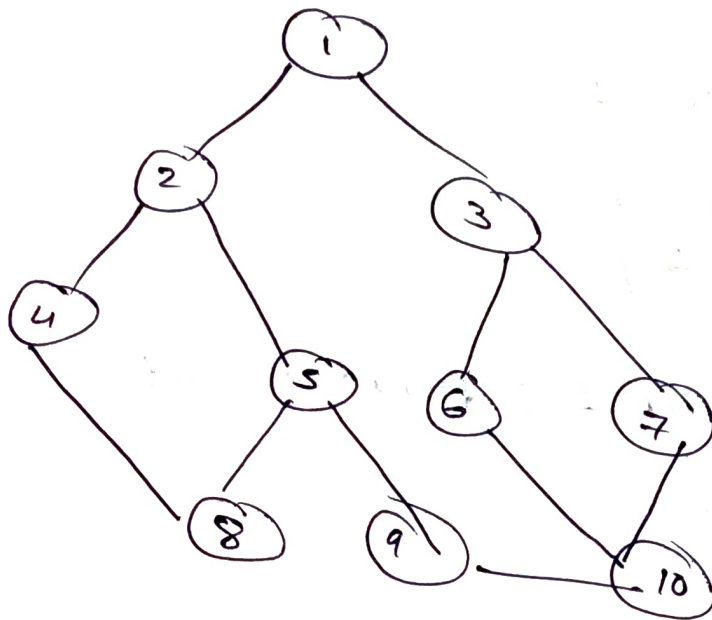
H	F	H	H	H
---	---	---	---	---

O/P: A B C D E F G H

step 17 : dequeue front

X	X	X	X
---	---	---	---

3



Adjacency List

1: [2,3]

2: [1,4,5]

3: [1,6,7]

4: [2,8]

5: [8,9,2]

6: [3,9,10]

7: [3,10]

8: [4,5,9]

9: [5,6,8,10]

10: [6,7,9]

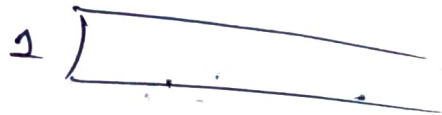
visited

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

step 1 : Enqueue 1



step 2 : visited 1 & dequeue front



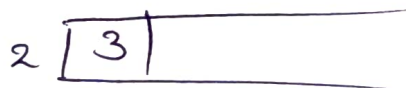
O/P : 1

step 3 : ~~enqueue~~ Enqueue Unvisited neighbours

of 1 $\rightarrow$ 2, 3



step 4 : visited 2 & Dequeue front



O/P : 1 2

step 5 : Enqueue Unvisited neighbours

of 2 $\rightarrow$ 4, 5



step 6: visited 3 & Dequeue front

3 | 4 | 5

O/P: 1 2 3

step 7: Enqueue Unvisited neighbours of 3 $\rightarrow$ 6, 7

4 | 5 | 6 | 7

step 8: visited 4 & Dequeue front

4 | 5 | 6 | 7

O/P: 1 2 3 4

step 9: Enqueue unvisited neighbours of 4 $\rightarrow$ 8

5 | 6 | 7 | 8

step 10: visited 5 & Dequeue front

5 | 6 | 7 | 8

O/P: 1 2 3 4 5

step 11: Enqueue unvisited neighbours of

5 $\rightarrow$ 8, 9

6 | 7 | 8 | 8 | 9

step 12: visited 6 & Dequeue front

6 | 7 | 8 | 8 | 9

O/P: 1 2 3 4 5 6

step 13: Enqueue unvisited neighbours of
6 $\rightarrow$ 9, 10

7	8	8	9	9	10
---	---	---	---	---	----

step 14: visited 7 & dequeue 7

7	8	8	9	9	10
---	---	---	---	---	----

O/P: 1 2 3 4 5 6 7

step 15: Enqueue unvisited neighbours

of 7 $\rightarrow$ 10

8	8	9	9	10	10
---	---	---	---	----	----

step 16: visited 8 & dequeue 8

O/P: 1 2 3 4 5 6 7 8

8	8	9	9	10	10
---	---	---	---	----	----

step 17: Enqueue unvisited neighbours

of 8 $\rightarrow$ 9

8	9	9	10	10	9
---	---	---	----	----	---

step 18: Since 8 is visited dequeue it

9 | 9 | 10 | 10 | 9

step 19: visited 9 & dequeued it

9 9 | 10 | 10 | 9

o/p: 1 2 3 4 5 6 7 8 9

step 20: Enqueue unvisited ~~ne~~ neighbours

of 9 → 10

9 | 10 | 10 | 9 | 10

step 21: Since 9 is visited dequeue it

9 10 | 10 | 9 | 10

step 22: ~~Enqueue unvisited neighbours~~

visited 10 & dequeue 10

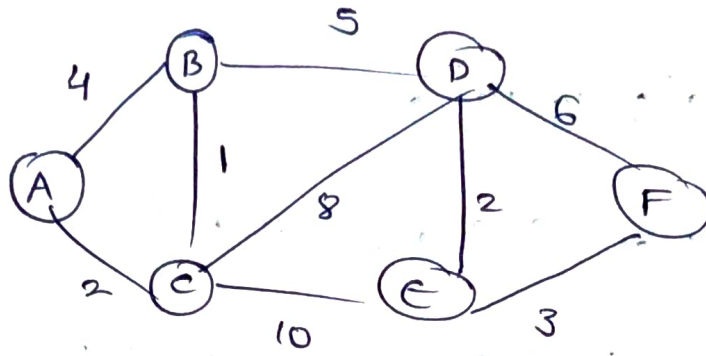
10 10 | 9 | 10

o/p: 1 2 3 4 5 6 7 8 9 10

step 23: Dequeue front

10 | 9 | 10

Dijkstra's Algorithm



<u>Node</u>	<u>Cost</u>	<u>previous</u>
A	0	
B	∞ 4	A
C	∞ 2	A
D	∞ 8	B
E	∞ 10	D
F	∞ 13	E

(A, 0)	(B, 3)	(C, 2)	(D, 8)	(E, 10)	(F, 13)
(A, 0)	(B, 4)	(C, 2)	(D, 10)	(E, 10)	(F, 14)

A → B $d(A) = 0$

$$d(A) + d(A, B) < d(B)$$

$$0 + 4 < \infty$$

4 < ∞ ✓ update

A → x

$$0 + 2 < \infty$$

2 < ∞ ✓ update

c → A

$$2 + 2 < 0 \times$$

don't update

c → B

$$4 + 1 < 4$$

3 < 4 update

c → D

$$2 + 8 < \infty$$

10 < ∞ ✓ update

B → A d(B) = 3

$$3 + 4 < 0$$

7 < 0 × don't update

B → c

$$3 + 1 < 2$$

4 < 2 (×) don't update

$$\underline{B \rightarrow D}$$

$$3 + 5 < 10$$

$$8 < 10 \checkmark \text{ update}$$

==

$$\underline{D \rightarrow B} \quad d(D) = 8$$

$$8 + 5 < 3$$

$$10 < 3 \text{ (X) don't update}$$

$$\underline{D \rightarrow C}$$

$$8 + 8 < 2$$

$$16 < 2 \text{ (X) don't update}$$

$$\underline{D \rightarrow E}$$

$$8 + 2 < \infty$$

$$10 < \infty \checkmark \text{ update}$$

$$\underline{D \rightarrow F}$$

$$8 + 6 < \infty$$

$$14 < \infty \checkmark \text{ update}$$

==

$$\underline{E \rightarrow C} \quad d(E) = 10$$

$$10 + 10 < 2$$

$$20 < 2 \text{ (X) don't update}$$

E → D

$$10 + 2 < 8$$

$12 < 8$ (X) don't update

E → F

$$10 + 3 < 14$$

$13 < 14$ ✓ update

=====

F → D $d(F) = 13$

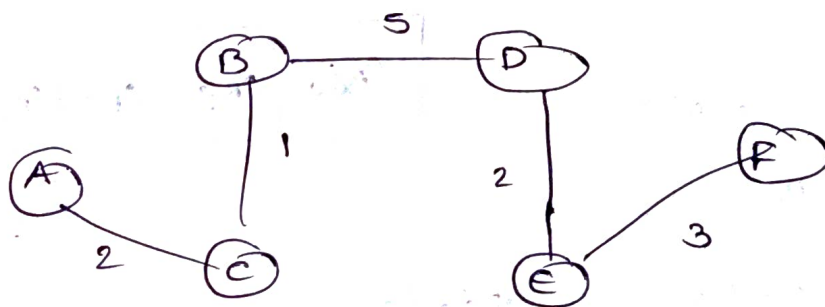
$13 + 6 < 8$ (X) don't update

F → G

$$13 + 3 < 10$$

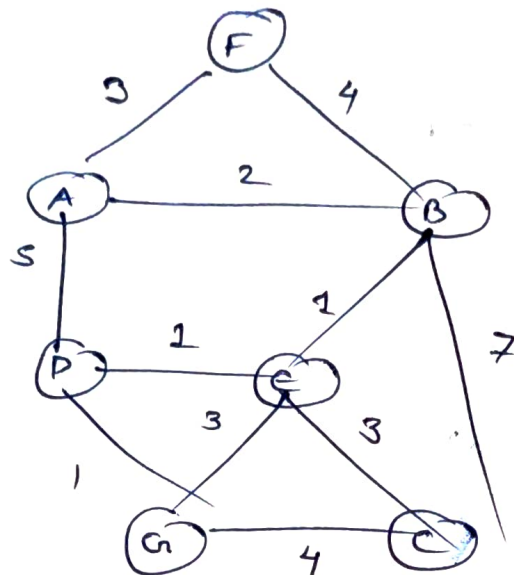
$16 < 10$ (X) don't update

shortest distance ! A → F



$$2 + 1 + 5 + 2 + 3 = 13$$

3



Node	Distance	prev
A	0	
B	∞ 2	A
C	∞ 6	B, E
D	∞ 4	A, E
E	∞ 3	B
F	∞ 3	A
G	∞ 5	E, D

~~∞~~ A $d(A) = 0$

A → B

$0 + 2 < ∞$ ✓ update

A → F

$0 + 3 < ∞$ ✓ update

(A, 0)	(B, 2)	(D, 5)	(F, 3)	(E, 3)	(G, 5)
					G, 5
					(G, 5)

$$\underline{A \rightarrow D}$$

$$0+5 < \infty \checkmark \text{ update}$$

at B

$$\underline{B \rightarrow F}$$

$$2+4 < 3 \text{ (X)}$$

$$\underline{B \rightarrow A}$$

$$2+2 < 0 \text{ X}$$

$$\underline{B \rightarrow E}$$

$$2+1 < \infty \checkmark$$

$$\underline{B \rightarrow C}$$

$$2+7 < \infty$$

$$9 < \infty \checkmark$$

At E

$$\underline{E \rightarrow B}$$

$$3+1 < 5$$

$$\underline{E \rightarrow G}$$

$$3+3 < \infty \checkmark$$

$$\underline{E \rightarrow C}$$

$$3+3 < 9$$

$$6 < 9 \checkmark$$

$$\underline{A \vdash F} \quad d(F)=3$$

$$\underline{F \rightarrow A}$$

$$3+3 < 0 \quad (\times)$$

$$\underline{F \rightarrow B}$$

$$3+4 < 2 \quad (\times)$$

$$\underline{A \vdash D} \quad d(D)=4$$

$$\underline{D \rightarrow A}$$

$$4+5 < 0 \quad (\times)$$

$$\underline{D \rightarrow E}$$

$$4+1 < 3 \quad (\times)$$

$$\underline{D \rightarrow G}$$

$$4+1 < 6 \quad \checkmark$$

$$\underline{A \vdash G} \quad d(G)=5$$

$$\underline{G \rightarrow D}$$

$$5+1 < 4 \quad (\times)$$

$$\underline{G \rightarrow E}$$

$$5+3 < 3 \quad (\times)$$

$$\underline{G \rightarrow C}$$

$$5+4 < 6 \quad (\times)$$

A to C:

C to B

$$6 + 7 < 2 \text{ (X)}$$

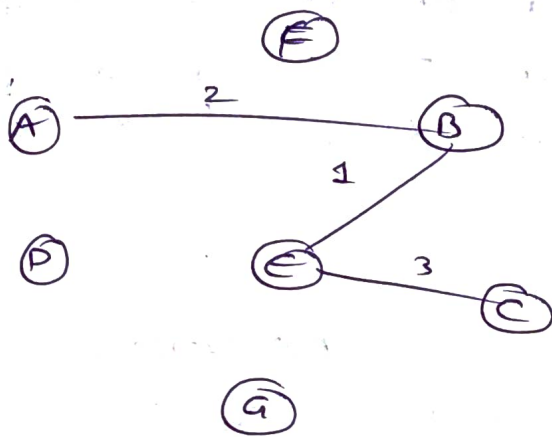
C to E

$$6 + 3 < 3 \text{ (X)}$$

C to G

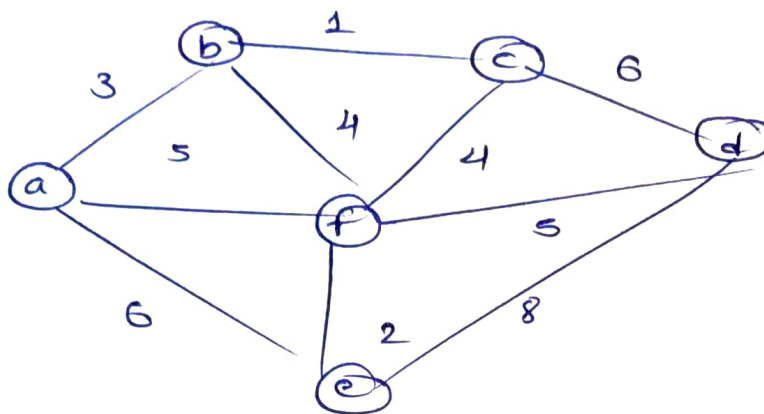
$$6 + 5 < 5 \text{ (X)}$$

Shortest distance from A to C



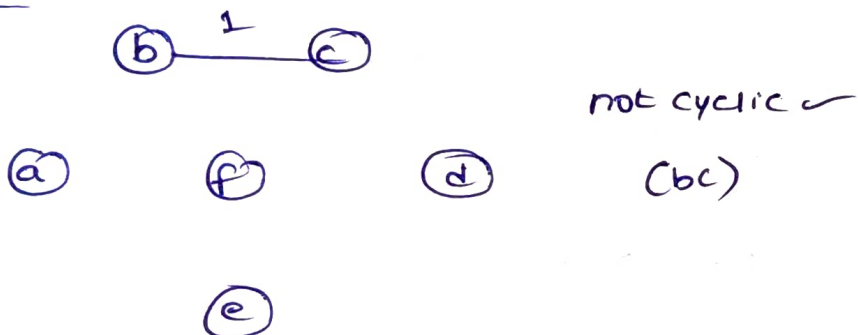
$$2 + 1 + 3 = 6$$

Kruskal Algorithm

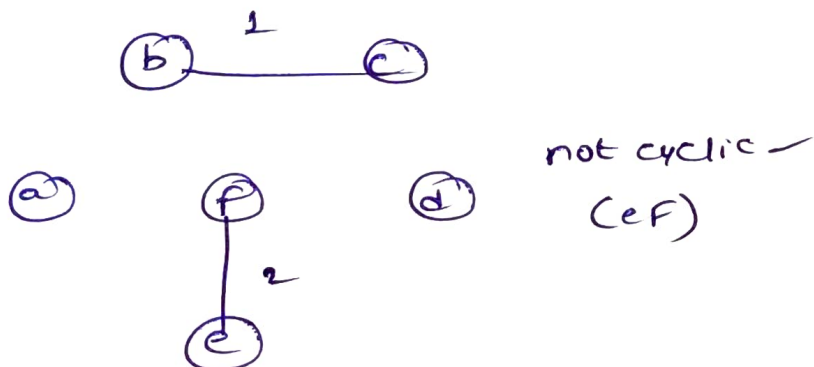


bc	ef	ab	bf	cf	af	df	ae	cd	de
1	2	3	4	4	5	5	6	6	8
✓	✓	✓	✓	✗	✗	✓	✗	✗	✗

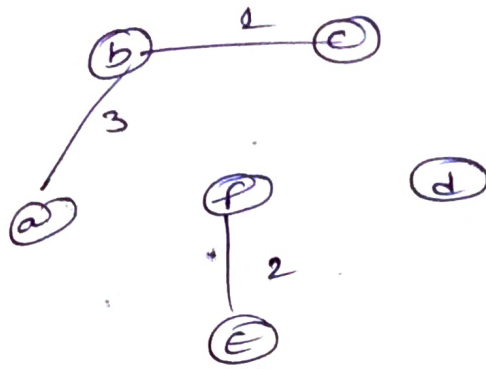
step 1



step 2

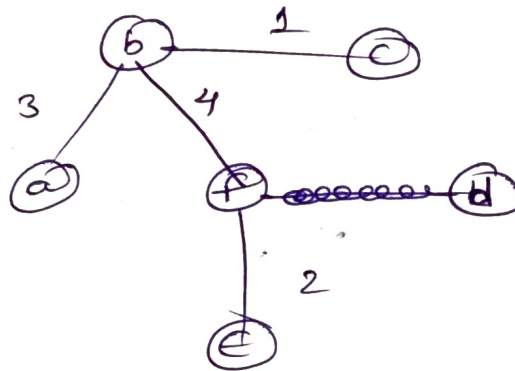


steps



not cyclic
(a,b)

steps

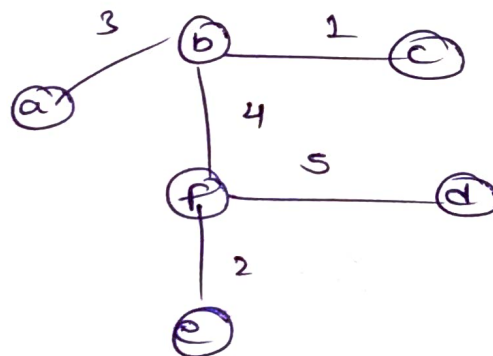


not cyclic
(b,f)

steps: cf forms cycle so leave it

steps: af forms cycle so leave it

steps (df)

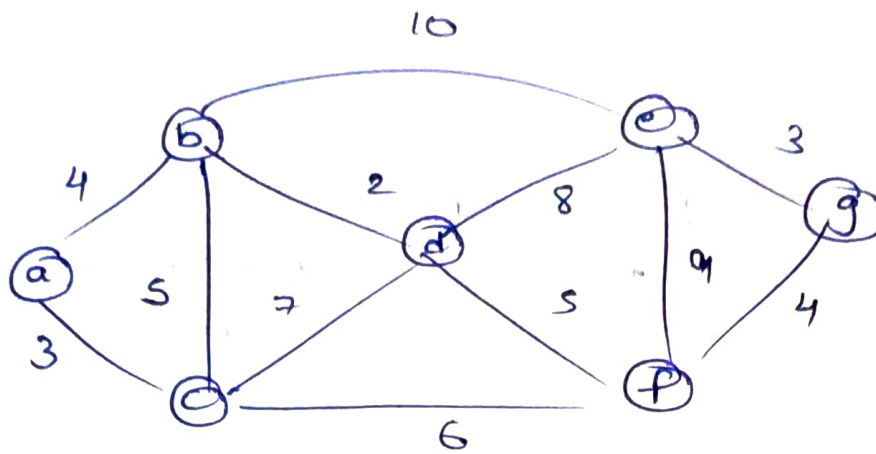


steps: ~~cd~~ ~~cd~~ ae, cd, de forms cycle so leave them

minium spanning tree weight

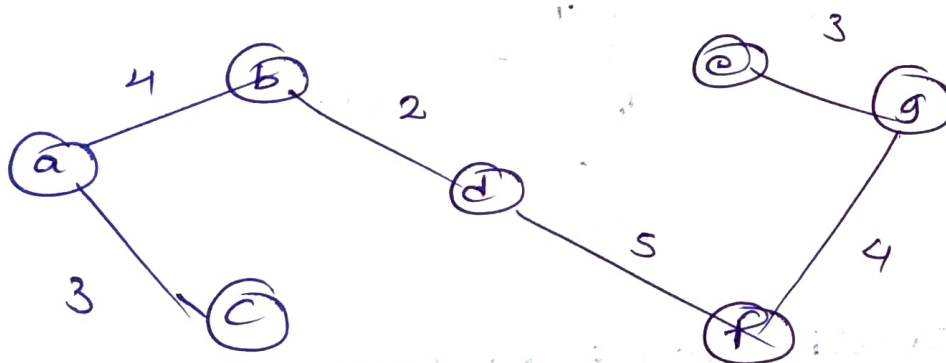
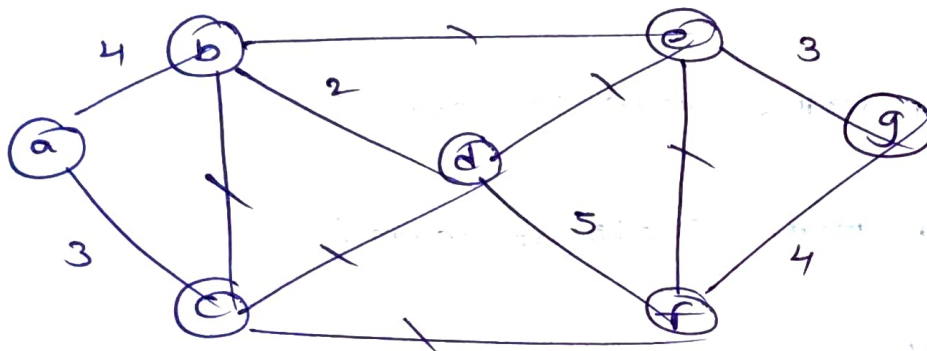
$$= 1 + 2 + 3 + 4 + 5$$

$$= 15$$

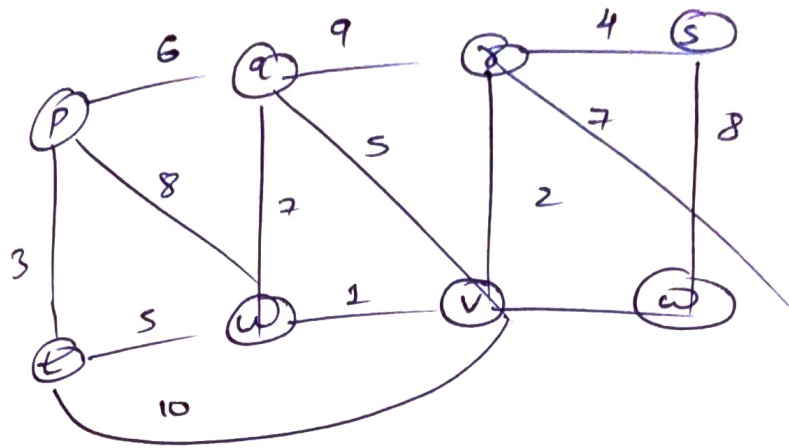


~~ab~~ ~~ac~~ ~~bc~~

bd	ac	eg	ab	fg	bc	df	cf	cd	de	ef	be
2	3	3	4	4	5	5	6	7	8	9	10
✓	✓	✓	✓	✓	X	✓	X	X	X	X	X

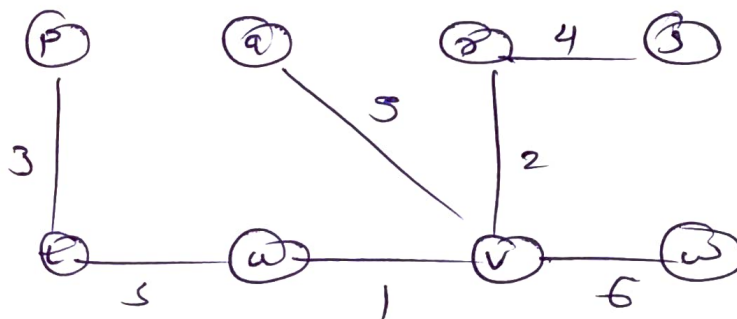
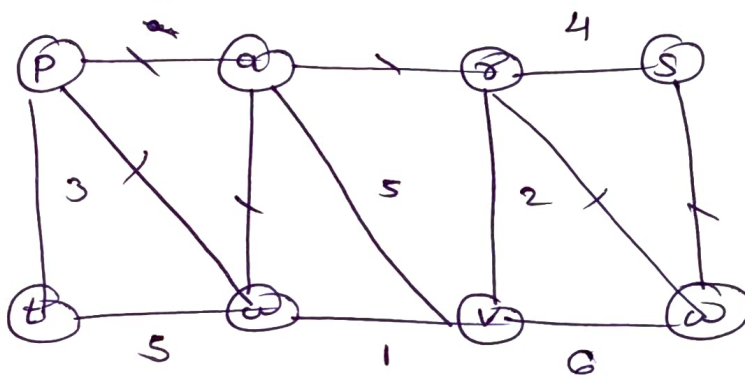


minimum spanning Time = $2 + 3 + 3 + 4 + 4 + 5$
 $= 21$



uv rv pt rs qv tu pq vw qu rw pu sw
 $\underline{1}$ $\underline{2}$ $\underline{3}$ $\underline{4}$ $\underline{5}$ 5 6 6 7 7 8 8

qr $+v$
 9 10



$$MST = 1 + 2 + 3 + 4 + 5 + 5 + 6$$

$$= 26,$$